DTC INSPECTION [LASER SENSOR]

id0402c4985000

1. Connect the M-MDS to the DLC-2.
2. After the vehicle is identified, select the following items from the initialization screen of the M-MDS.
   (1) Select “Self Test”.
   (2) Select “All CMDTCs”.
3. Verify the DTC according to the directions on the screen.
   - If any DTCs are displayed, perform troubleshooting according to the corresponding DTC inspection. (See DTC TABLE [LASER SENSOR].)

   Note
   - Snapshot data appears at the top of the help screen when the displayed DTC is selected.
   - Snapshot data stores the currently detected DTC data.

4. After completion of repairs, clear all DTCs stored in the module. (See CLEARING DTC [LASER SENSOR].)

Snapshot Data Table

Note
- The laser sensor stores the following two types of information when a DTC is detected and displays snapshot data in the M-MDS
  — Vehicle information detected by laser sensor
  — Vehicle information detected by instrument cluster and received via CAN communication
- Refer to PID monitor table for confirm the laser sensor system operation status while the laser sensor does not store the DTC. (See PID/DATA MONITOR TABLE [LASER SENSOR].)
- Snapshot data items are not displayed, according to detected DTC.

—: Not applicable

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
AAT	°C, °F	Ambient air temperature	—	—
APP_STATUS	Accelerator Pedal Off/ Under 20%/ Over 20%/ FAIL	Accelerator pedal position	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error	Configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/ Over 80 degrees C/ FAIL	Engine coolant temperature status	—	—
IC_VPWR	V	Instrument cluster power supply	• The laser sensor constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the M-MDS.	VPWR*1

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
IG-ON_TIMER	hh:mm:ss*2	Elapsed time since ignition was switched ON **Note** • The instrument cluster records the elapsed time since the ignition was switched ON.	• The laser sensor constantly receives the elapsed time since the ignition was switched ON sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the elapsed time since the ignition was switched ON when the DTC was detected, and it is displayed in the M-MDS.	—
PWR_MODE_KEY	Key Out/Key Recently Out/Key Approved (Position 0)/Post Accessory (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 1)/Ignition On (Position 2)/Running (Position 2)/ Running - Starting In Progress (Position 2)/ Crank (Position 3)	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time within 3 s since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time within 3 s since ignition was switched ON • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The laser sensor constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—
RPM_STATUS	Engine Stop/ Under 1500 rpm/ Over 1500 rpm/ FAIL	Engine RPM status	• The laser sensor constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	TACHOMTR*1

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
SHIFT_STATUS	P/N D/ R/ FAIL	Shift position status	• The laser sensor constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the selector lever position when the DTC was detected, and it is displayed in the M-MDS.	—
TOTAL_DIST	km, ft, mi	Accumulated total traveled distance from completion of vehicle until laser sensor detects DTC (Odometer value in instrument cluster)	The distance traveled when the laser sensor detected a DTC can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snap shot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*1	Accumulated total elapsed time since vehicle completion until laser sensor detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time when the laser sensor detected a DTC can be calculated by performing the following procedure. 1. Verify the PID item TOTAL_TIME of the instrument cluster. 2. Verify the snap shot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*1
VPWR	V	Power supply	—	VPWR_IG1
VSPD_STATUS	Stop/ 0 - 10 km/h/ Over 10 km/h/ FAIL	Vehicle speed status	• The laser sensor constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the laser sensor records the vehicle speed when the DTC was detected, and it is displayed in the M-MDS.	SPEEDOMTR*1

*1 : PID for instrument cluster (See .)
*2 : Fractions of seconds may be indicated after the decimal point